

1. A pharmaceutical company planned to release a new sustain theophylline tablet. Based on ICH recommendation protocols for stability study, the company is on the verge to start the study.

What is the most appropriate recommendation to this company?

- (A) Stability study commitment is good to apply
- (B) Six month real-time stability data is enough
- (C) Accelerated stability is enough for decision
- (D) Real-time stability study is not mandatory

Answer: A

2. A Pharmaceutical factory manufactured a new drug substance in tablet form. The drug substance is prone to decomposition by moisture and light. After manufacturing, the pharmacist selected and sealed the product in appropriate primary container for marketing.

What is the most appropriate primary packaging material that the pharmacist should use?

- (A) Blister with desiccant packet
- (B) Opaque glass bottle without desiccant packet
- (C) Tight plastic container without desiccant packet
- (D) Amber-colored container with desiccant packet

Answer: D

3. In capsule production line, a production pharmacist is manufacturing capsule containing hydroscopic material. After the capsules are produced, the available packaging materials are gelatin capsule, capsule of vegetable origin, polyvinyl alcohol and hypromellose capsules.

What is the most appropriate packaging material for this product?

- (A) Gelatin
- (B) Hypromellose
- (C) Polyvinyl alcohol
- (D) Vegetable origin

Answer: B

4. A manufacturing pharmacist working in a pharmaceutical company manufactured mass of enteric coated granules of omeprazole enclosed in capsule. He has packed and labeled the product and is waiting for product release.

What is the most likely requirement that the pharmacist is waiting for?

- (A) Preparation of store
- (B) Request from the market
- (C) In-process quality control test result
- (D) Approval from quality control unit

Answer: D

5. A prescription came from a dermatology department of a public hospital to a community pharmacy. The prescription orders the pharmacist to prepare suitable semisolid dosage form for a patient having highly secretive wound and inflammation at a hairy skin part of his leg.

What is the most appropriate dosage form in this scenario?

- (A) Pastes
- (B) Creams
- (C) Ointment
- (D) Plaster

Answer: B

6. A pharmaceutical technologist wants to prepare an O/W type emulsion cream. While preparing the oil and water phases, she referred to Griffin's HLB numerical scale for selecting suitable emulsifying agent.

What is the most suitable emulsifier for the cream?

- (A) 3 to 8
- (B) 7 to 9
- (C) 9 to 16
- (D) 16 to 20

Answer: C

7. A pharmaceutical factory manufactured a batch of miconazole nitrate cream (2%). The product is waiting for specified in-process quality control test results.

What should the manufacturing pharmacists do until the requested tests are complete?

- (A) Filling
- (B) Storing
- (C) Sterilizing
- (D) Packaging

Answer: B

8. A pharmacist working in the research and development department of a given pharmaceutical industry planned to prepare water in oil emulsion.

What should be done to meet the above plan?

- (A) Adding oil phase to water phase
- (B) Dissolving drug in the aqueous phase
- (C) Using high volume of water phase than oil phase
- (D) Dissolving sweeteners and flavorant in aqueous phase

Answer: B

9. A pharmaceutical formulator should make effort to avoid certain rheological characteristics and in-place some rheological characteristics during designing & developing suspension formulation to improve its physical stability.

Which rheological characteristic is most likely introduced into suspension preparation to improve its physical stability?

- (A) Pseudo plastic flow
- (B) Dilatant flow
- (C) Newtonian flow
- (D) Plastic flow

Answer: D

10. After manufacturing, a pharmacist sterilized the final aqueous solution form of a drug product in sealed containers using high pressure of saturated hot steam.

What was the most appropriate sterilization technique that the pharmacist used?

- (A) Gas
- (B) Radiation
- (C) Filtration
- (D) Autoclaving

Answer: D

11. A pharmacist working in a pharmaceutical industry research and development unit designed acetylcholine chloride in powder form for intra-ocular injection. Finally, the powder is packed in multiple-dose container and dispensed with solvent for reconstitution.

What is the most appropriate container the pharmacist should use for this product?

- (A) Vial
- (B) Ampoule
- (C) Canister
- (D) Prefilled syringe

Answer: A

12. A manufacturing industry produces heat stable sterile drug products by applying compatible highly reproducible and generally readily validated lethal treatment method.

What method is most likely important for such industry?

- (A) Aseptic processing
- (B) Terminal sterilization
- (C) Quality control testing
- (D) Process simulation testing

Answer: B

13. A warehouse personnel is assigned to manage a proper storage of biological products to maintain shelf life of the products.

What is the most appropriate handling condition for these products?

- (A) Storing the products with other products in the cold chain
- (B) Placing expired products at one compartment of the cold chain
- (C) Placing the products near to the cooling plate of the cold chain

(D) Allowing sufficient space in the cold chain for air to circulate
Answer: D

14. A nurse comes to a community pharmacy with a prescription for ipratropium bromide aerosol solution. The nurse requires the medication to give a home care for her patient. The pharmacist advised the nurse on the requirement of special storage condition.

What is the most appropriate storage condition the pharmacist should recommend?

- (A) Keep it away from excess heat
- (B) Place at freezing temperature
- (C) Put in area with access to light
- (D) Store in area with excess humidity

Answer: A

15. The GMP inspection team observed sliding door between ancillary area and production room of a manufacturing company. The team recorded this as a non-conformance works to be corrected.

What is the most likely reason for the team to record the presence of sliding door as nonconformance?

- (A) To Prevent the pressure differential from pushing the door open
- (B) To prevent pressure differential from disturbance of air flow
- (C) To prevent the accumulation dust in sliding grooves.
- (D) To prevent any person to easily open.

Answer: A

16. An EFDA inspection team inspected a pharmaceutical manufacturer and recorded the following observations among others. The production premises had ceilings and walls that do not allow air passage, multiple sharp edges and corners, close fitting doors, and sealed light fittings. At the end of the inspection and after completing every required task, the team finally concluded that the premises were not fit for pharmaceutical production.

What feature of the premises most likely contributed for the conclusion made by the team?

- (A) Sealed light fittings
- (B) Close fitting doors
- (C) Edges and corners
- (D) Ceiling and walls

Answer: C

17. A local pharmaceutical company has got a manufacturing certificate of competency (license) from FDA. Due to shortage of hard currency, the company failed to start business. After one year of license issuing, inspection team from EFDA approached the company and came up with a decision of license suspension.

- What is the most likely reason for the team to reach at such a decision?
- (A) The company failed to start business within 6 months
 - (B) The company has no GMP certificate
 - (C) The license is not renewed
 - (D) Service fee not paid

Answer: A

18. A pharmaceutical manufacturer licensed by the EFDA is normally expected to be periodically inspected by a team from the authority.

What is the main purpose of such periodic inspection?

- (A) To revoke the license of the manufacturer
- (B) To issue manufacturing certificate of competence
- (C) To find faults and malpractices in the establishment
- (D) To issue GMP certificate in accordance with EFDA directives

Answer: D

19. In a pharmaceutical manufacturing industry, an annual internal quality audit was conducted and during the inspection, different non-conformity (NC) activities were detected. Most of those detected NC activities were found to be major NC. The audit team communicated these NC with the quality assurance manager of the company.

What should be the most appropriate next step of the quality assurance manager?

- (A) Preventive action
- (B) Corrective action
- (C) Management review
- (D) External audit

Answer: C

20. An analyst in quality control laboratory received a dissolution test request for tablet sample from production line. Before performing the analysis, the analyst identified the Bio pharmaceuticals Classification (BSC) of the product as BSC-class I. Once identified, the BSC class of the drug, the analyst conducted the dissolution test and recorded the amount dissolves as Q at dissolution time of 30 minutes.

What is the minimum amount of the drug (Q) expected to be dissolved at 30 minutes in this case?

- (A) $Q \geq 50\%$ of the labeled amount
- (B) $Q \geq 60\%$ of the labeled amount
- (C) $Q \geq 70\%$ of the labeled amount
- (D) $Q \geq 80\%$ of the labeled amount

Answer: D

21. A variety of quality control tests are performed by FDA as part of the process of granting marketing authorization to pharmaceutical products entering the market. The FDA wants to ensure that each units of a pharmaceutical formulation contain reasonable amount of the active ingredient.

What type of quality control tests are the most appropriate in this case?

- (A) Dissolution and physical strength tests
- (B) Assays and dosage uniformity tests
- (C) Disintegration and brittleness tests
- (D) Elasticity and water content tests

Answer: B

22. The WHO has established certain minimum regulatory requirements for the registration of herbal traditional medicines in its guideline. Among these requirements are quality control parameters.

What quality control parameters are used as part of purity tests on these types of products?

- (A) Description of the process of manufacture
- (B) Quantification of the active ingredient
- (C) Quantity and type of excipients
- (D) Pesticide residues

Answer: D

23. A pharmaceutical promotion expert (medical representative) working in Ethiopia for the past five years in private pharmaceuticals importing company is being assessed whether he/she abides to EFDA promotion norms.

What type of acceptable behavior should the expert show in this scenario?

- (A) Switching between jobs/employment without notifying the authority
- (B) Presenting expensive promotional items as gifts to prescribers
- (C) Ignoring any unethical practice pattern by the hiring company
- (D) Offering free medical samples to health professionals

Answer: D

24. An Ethiopian pharmacy professional studied a Bachelor of degree in pharmacy in a foreign country. After receiving a professional license from the country's authority, the professional was not able to get a job in that country for about four months and wanted to return to Ethiopia to practice her profession.

What is the most appropriate initial step before starting a job in Ethiopia?

- (A) Complete a one year internship program
- (B) Get a new professional license in Ethiopia
- (C) Get a temporary professional license from Ethiopia
- (D) Produce evidence for equivalence of qualification

Answer: D

29. A 35-year-old severe cancer patient is admitted to a specialized hospital. The patient's pain is serious and stressing despite taking several pain relieving medications. While the pharmacist was rounding in the ward with other physicians, the patient requested the team that the medications are not working well and assist him to die and get relief from his pain.

What is the legal implication of this scenario in Ethiopia?

- (A) Assisted suicide is allowed when the patient is autonomous
- (B) Assisted suicide is not allowed according to Ethiopia's law
- (C) Assisted suicide is allowed for patients with life-threatening illnesses
- (D) Assisted suicide is allowed when the patient and family members allow

Answer: B

30. A pharmacist is working in a hospital pharmacy of a certain research center and is classifying controlled substances based on their schedule on the controlled substance acts. In doing so, the pharmacist saw a product with a label of "for research purpose only".

What is the most likely substance the pharmacist observed in this case?

- (A) Heroin
- (B) Diazepam
- (C) Codeine
- (D) Morphine

Answer: A

31. A study is conducted to assess the effect of provision of Folic Acid to pregnant women on the birth weight of their babies.

What type of results is the researcher most interested in this scenario?

- (A) Qualitative results
- (B) Descriptive results
- (C) Quantitative results
- (D) Both qualitative and quantitative results

Answer: C

32. A male researcher wanted to state the objective of his research entitled 'antiepileptic medication adherence among epileptic patients in 'X' hospital'.

Which of the following is the correct way of stating his objective?

- (A) To appreciate antiepileptic medication adherence
- (B) To understand antiepileptic medication adherence
- (C) To determine antiepileptic medication adherence
- (D) To believe antiepileptic medication adherence

Answer: C

33. A researcher is highly interested to conduct a study entitled "Does the use of antipsychotic medications on patients with major mental disorders associated with the risk of obesity?" using a medical chart of patients with and without obesity.

Which design is most appropriate for this study?

- (A) Retrospective cohort
- (B) Prospective cohort
- (C) Cross sectional
- (D) Case-control

Answer: D

34. A researcher wanted to conduct a cross-sectional study on the prevalence of 'khat' chewing and its associated factors among high school students in Ethiopia.

What is the most appropriate sampling method to be used in this case?

- (A) Simple random sampling
- (B) Stratified random sampling
- (C) Sequential random sampling
- (D) Systematic random sampling

Answer: B

35. A graduate student proposed to conduct a study on the prevalence of non-adherence to ART medications and its associated factors among 400 patients at an ART clinic of a certain hospital. The student prepared a self-administered interview guide after an extensive literature review.

What is the most appropriate next step in the above case?

- (A) Conducting a pre-test on 20 patients in different hospitals
- (B) Conducting the study on 440 patients in the hospital 'X'
- (C) Changing the questionnaire into interview based type
- (D) Entering the data on the suitable statistical software

Answer: A

36. In a given dataset, an extremely low or high observations ends up with "skewness" of a distribution curve. Hence, the mean value tends to shift towards those scores. When majority of scores are at the left end of the curve and few extreme large scores are scattered at the right end, a skewed distribution will occur.

What is the most likely type of skewed distribution in the above case?

- (A) Unimodal skewed distribution
- (B) Centrally skewed distribution
- (C) Positively skewed distribution
- (D) Symmetrical skewed distribution

Answer: C

37. A measure of variation or dispersion of given data sets determined to be the difference between the highest and smallest observation, known as crudest measure of dispersion and cannot be usually employed for comparing variability of two distributions expressed in different units.

What is the most likely measure of variance the above paragraph refers to?

- (A) Inter quartile range
- (B) Standard deviation
- (C) Range
- (D) Quartile

Answer: C

38. In a study conducted on the association between drinking coffee and heart failure, the relative risk of the association was found to be 0.6 with a confidence interval of 0.4-0.8.

What is the most appropriate interpretation of this finding?

- (A) A valid comparison is not possible with the given case
- (B) The risk of heart failure is lower among coffee drinkers
- (C) The risk of heart failure is higher among coffee drinkers
- (D) There is no association between coffee drinking and heart failure

Answer: B

39. A field epidemiologist wanted to calculate deaths due to car accident in a certain city and started to consult a colleague on the appropriate epidemiologic outcome measure to be applied for his question.

What is the most appropriate measure to be used in this case?

- (A) Disability adjusted life year
- (B) Case fatality rate
- (C) Proportionate mortality rate
- (D) Years of potential life lost

Answer: B

40. An organization of a research paper refers to the structure and/or the sequence in which you present each type of information. The scientific report should have distinctive and clearly evident component parts. In one of the components, you will be expected to describe in detail how and with what means you accomplished your project.

Which component of a research paper most likely describes the above expectation?

- (A) Abstract
- (B) Introduction
- (C) Statement of the problem
- (D) Methods and Materials

Answer: D

41. A hospital pharmacy head is developing the 2021/2014 E.C. fiscal year's budget plan by categorizing into four groups; internal process, finance, human resources and capacity building.

What is the most likely category of the planning?

- (A) Strategic planning

- (B) Program planning
- (C) Operational planning
- (D) Project planning

Answer: C

42. A hospital management team has solved a prolonged loss of medicines through expiry and theft. The team concentrated on the causes to understand the wider range of the problem.

Which building block of a health system is addressed by the above team?

- (A) Health work force
- (B) Information system
- (C) Services delivery system
- (D) Leadership and governance

Answer: D

43. A pharmacy case team was established to develop monitoring and evaluation plan for reproductive health commodities supply in a certain region.

What is the most appropriate first step to be conducted by the team?

- (A) Assessment
- (B) Develop goal
- (C) Set objectives
- (D) Prioritize intervention

Answer: A

44. The national drug policy of the transitional government of Ethiopia was ratified in 1993. It has six objectives and thirteen strategies to accomplish those objectives. One of the strategies is "the government shall make all efforts to make sure that the public gets drugs that are within its means".

What is the most likely strategy that the above paragraph refers to?

- (A) Research and development
- (B) Drug administration and control
- (C) Drug information and promotion
- (D) Drug pricing

Answer: D

45. A drug and therapeutic committee wants to compare drug 'X' which is manufactured by Pfizer pharmaceutical company and, drug 'Y' which is manufactured by Ethiopian pharmaceutical company. Evidence shows that the two drugs have similar bioequivalence.

What is the most appropriate pharmaco-economic analysis to use for a decision?

- (A) Cost benefit analysis
- (B) Cost effectiveness analysis
- (C) Cost minimization analysis
- (D) Cost utility analysis

Answer: C

46. A junior pharmacist was assigned to be head of Pharmacy Unit of a newly opened health center located in one of the 'Woredas'. There was a previous health center that started providing health care service two years back. The pharmacist needed to estimate the total quantity and cost of commodities necessary for the new health center.

What is the most appropriate data to take from the first health center for the quantification?

- (A) Consumption data
- (B) Service data
- (C) Morbidity data
- (D) Disposal data

Answer: B

47. Pharmaceutical's procurement team must ensure that products meet regulatory and shipping/packaging requirements. Besides, it make sure the products are manufactured at facilities that meet GMP certification requirements.

What is the most likely activity of the team in this scenario?

- (A) Monitor contract performance
- (B) Award contract
- (C) Supplier selection
- (D) Specification

Answer: D

48. A head of a pharmacy case team of a primary hospital requested financial assistance from a local NGO and has the probability of obtaining the assistance from the NGO. As a matter of procedure, the NGO asked for quantification of the financial request before releasing the assistance.

What is the most appropriate quantification method to use for the scenario?

- (A) Consumption
- (B) Morbidity
- (C) Proxy-consumption
- (D) Service level projection

Answer: D

49. A hospital purchased 100,000 units IV infusion of Metronidazole for the 2014 fiscal year. However, 2.5% was lost due to damage during transportation, 2.5% was lost due to expiry and, 5% was lost due to theft. The remaining was consumed properly in the 2014 fiscal year. The consumption of the drug from 2014 is expected to be increased by 10% in the 2015 fiscal year (FY).

What is the most appropriate quantity of the same drug to be ordered in 2015 FY?

- (A) 90,000
- (B) 99,000

(C) 108,000

(D) 108,900

Answer: D

50. A pharmacist is working in a drug supply management unit of a hospital, which was established ten years back. The hospital mostly purchases its health commodities from local suppliers, unless there is an emergency stock-out. The hospital has a complete record of the purchase and utilization of pharmaceuticals. There was also a relatively uninterrupted supply trend of pharmaceuticals. Thus, the pharmacist decided to quantify an amount of pharmaceuticals for the next budget year.

What is the most appropriate quantification method to be used in this scenario?

(A) Proxy-consumption

(B) Consumption

(C) Morbidity

(D) Service level projection

Answer: B

51. In a certain hospital, Ceftriaxone injection is usually delivered after two weeks of ordering. A total of 12,000 vials of Ceftriaxone were consumed in the last four months and there are 15,000 vials left in the stock.

What is average monthly consumption of Ceftriaxone in this hospital?

(A) 2000

(B) 3000

(C) 4000

(D) 5000

Answer: B

52. A hospital pharmaceutical consumption data indicated that 60% of pharmaceuticals were fast moving, 25% were moderate moving and, the remaining were slow moving. Fast moving items cost 75% of the budget and moderate moving items cost 15% of the budget.

What is the most appropriate purchasing model for the slow moving pharmaceuticals?

(A) Perpetual

(B) Annual

(C) Scheduled

(D) Twice a year

Answer: B

53. A hospital drug and therapeutics committee wants to prepare an essential medicines list and started to identify the relevant factors to consider in selecting the medicines.

Which is the most relevant factor to be considered in this case?

- A) Availability in the international market
- B) The national disease pattern
- C) Unit prices of the medicines
- D) The local disease pattern

Answer: D

54. The Ethiopian Pharmaceutical supply Agency (EPSA) stocked out of Insulin 1500 IU. Because of the long lead time to purchase from foreign suppliers, the agency decided to procure 50000 Insulin Vials to be distributed to health facilities to cover the demand until the ordered drug is shipped. During the procurement process the agency bid and bargains three local pharmaceutical companies and one of the companies was selected to supply the medication.

What method of Procurement did EPSA use to acquire the medication?

- (A) Direct procurement method
- (B) Competitive negotiation method
- (C) Open tender procurement method
- (D) Closed tender procurement method

Answer: B

55. A pharmacy store manager working in a public health facility follows the Integrated Pharmaceuticals Logistics System of Ethiopia (IPLS)'s recommendation for managing the inventory. The pharmacist is triggered for placing an order of some essential pharmaceuticals to the maximum stock level.

What is the most likely stock level of the pharmaceuticals at present?

- (A) Adjustment Stock level
- (B) Safety stock level
- (C) Lead-time stock level
- (D) Minimum Stock level

Answer: D

56. A hospital store manager received the purchased pharmaceuticals and transcribed information from a purchase invoice into a receiving voucher. The transcribed information include; date, name of the supplier, invoice number, code of the product, description of the product, batch number, unit of measurement, quantity received, unit and total cost, expiry date and the name of individual that delivered the product.

What is the most likely receiving voucher used by the store manager?

- (A) Model 19
- (B) Model 20
- (C) Model 22
- (D) Model 06

Answer: A

57. A store manager is revising data that contain information about the quantity of goods moved from one storage facility to another (either between levels or within the health facility).

What is the most likely type of data the store manager is revising?

- (A) Pipeline data
- (B) Fleet management data
- (C) Issue data
- (D) Consumption data

Answer: C

58. A public health facility is usually stocked out of essential pharmaceuticals. Delay in deliveries and increased consumption and losses are among the reasons identified through assessment. Head of the pharmacy department decided to determine and hold stocks of pharmaceuticals to manage the stock out risk.

What is the most likely type of stock to be held by pharmacy department?

- (A) Lead time stock
- (B) Buffer stock
- (C) Review period stock
- (D) Minimum stock

Answer: B

59. A national formulary and therapeutic board approved the length of time from manufacturing pharmaceuticals to the final date that the pharmaceuticals can be safely used. The board accepted what manufactures specified as a reference.

What is the most likely nature of pharmaceuticals the above description refers to?

- (A) Shelf life
- (B) Expiration date
- (C) Storage duration
- (D) Pipeline duration

Answer: A

60. A pharmacist has purchased Oxytocin, Diazepam, Amoxicillin, and Anti-D from Ethiopian pharmaceutical supply agency.

What is the most appropriate shipment condition for the purchased drugs?

- (A) Cold box for Diazepam and Ant-D
- (B) Cold box for Oxytocin and Anti-D
- (C) Cold box for Diazepam and Oxytocin
- (D) Cold box for Amoxicillin and Diazepam

Answer: B

61. A pharmacist is given a responsibility for checking and approving shipments dispatched to health facilities. The pharmacist checked and found some cartons with voids because some of the products were taken out to ensure delivery of the right quantity.

What is the most appropriate action the pharmacist should take?

- (A) Fill the voids with packaging material
- (B) Seal the carton to reduce theft by movers
- (C) Transport the products without the carton
- (D) Transfer the product to a carton for another medicine

Answer: A

62. A client with a history of diazepam use for insomnia is experiencing craving symptoms, and consulted a community pharmacist for an advice on the possible management options.

What is the most appropriate initial action in this case?

- (A) Keeping a good sleep hygiene
- (B) Maintaining the same medication
- (C) Change to amitriptyline every other day
- (D) Change to a long acting benzodiazepines

Answer: A

63. A clinical trial was being conducted to test the effectiveness of a new anti-dandruff extract from *Aloe Vera*. At clinical stage, participants were randomly assigned to the experimental and control group.

What should the pharmacist recommend regarding the medication administered for the control group?

- (A) Use pre-existing anti-dandruffs
- (B) Administer purified water
- (C) Use the new drug at low dose
- (D) Leave with no medication

Answer: A

64. Ethical issues related to health can be macro or micro depending on the type of decision maker's involvement to address the ethical dilemma encountered during a service delivery. Macro ethical issues/dilemmas require involvement of legislative body of the country. On the other hand micro ethical issues could be handled at an individual/practitioner level.

Which ethical issue/dilemma is most likely to be handled by the healthcare practitioner?

- (A) Abortion
- (B) Euthanasia
- (C) Confidentiality
- (D) Artificial insemination

Answer: C

65. A patient from a psychiatry department is presented to OPD pharmacy requesting for an OTC anti-pain.

What is the most appropriate measure that the pharmacist should take?

(A) Dispense with ordinary prescription

(B) Request special prescription

(C) Dispense without a prescription

(D) Reject dispensing request

Answer: D

66. A private pharmacy owner applied to a drug regulatory authority to open pharmacy on the fourth floor of an apartment. The owner has fulfilled the rest requirements. However, the request could not get acceptance.

What is the most likely violated ethical principle which resulted in rejection?

(A) Justice

(B) Non-maleficence

(C) Autonomy

(D) Beneficence

Answer: A